GStar Assignment 2

Linear attention and Kimi Delta Attention

From Scratch

Efficient inference in frontier models

Alibaba Cloud

Qwen3.5/3.8/4

Gated DeltaNet (GDN)
+ full attention

Moonshot AI

Kimi K3

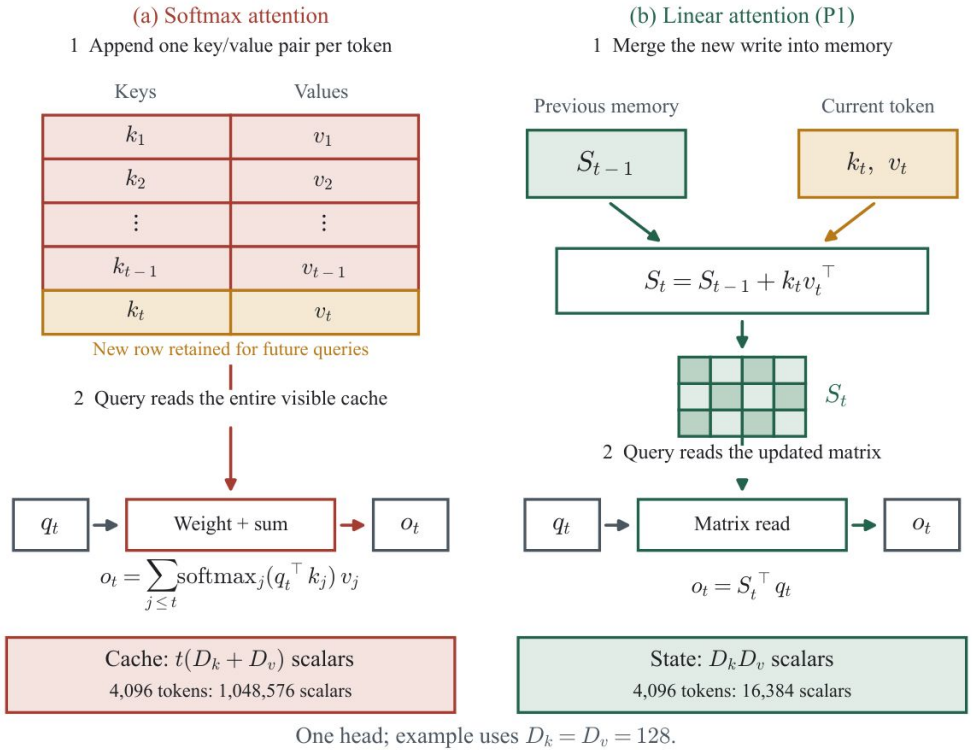
Kimi Delta Attention (KDA)
+ full attention (MLA)

NVIDIA

Nemotron 3 Super

Mamba-2 state space
+ full attention

Two ways to carry history



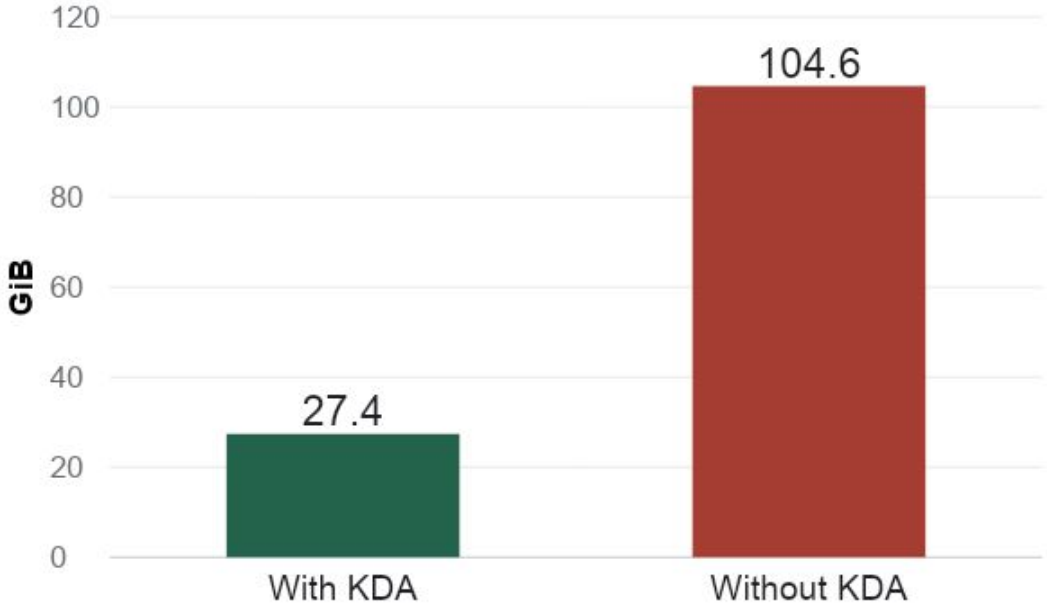
Token cache

Each new token adds a key and a value.

Matrix memory

Each new token changes the same-sized state.

Kimi K3 at one million tokens



74%

less attention-history storage

69 KDA + 24 MLA layers versus 93 MLA layers

One request at 2^{20} tokens. Calculated storage, before sharding. The all-MLA case is hypothetical.

BF16 MLA cache + FP32 KDA states. Excludes weights, convolution states and temporary buffers.

The nine-problem path

Matrix memory		Forgetting and correction		Kimi Delta Attention	
P1	PyTorch foundation	P4	Gated linear attention	P7	Gated DeltaNet
P2	Recurrent Triton	P5	Recurrent DeltaNet	P8	KDA recurrence + scores
P3	Chunked linear attention	P6	Chunked DeltaNet / WY	P9	Complete chunked KDA

Each stage gives the next stage a recurrence, a kernel pattern or a reusable primitive.

Implementation and correctness

Forward pass only

The checker compares outputs and final state$_v$against an independent recurrence.

https://github.com/newturing/GStar-Assignment-2-KDA

NEXT WEEK: (Bonus questions)

1. Implement the entire architecture and train on real data.
2. Implement new techniques: Multi Token Prediction and Speculative Decoding
3. Eval the trained model on real tasks